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Studierichting: Informatica
Oefeningenreeks 1C nr. 47.2

$$\begin{aligned} z^5 &= -1 \\ &= 1 \cdot \operatorname{cis} \pi \\ &= \operatorname{cis} \left(\frac{\pi + 2\pi k}{5} \right), \forall k \in \mathbb{N} \cap [0, 4]. \end{aligned}$$

Dus de geldige wortels zijn:

$$\begin{aligned} z_0 &= \operatorname{cis} \frac{\pi}{5} \\ z_1 &= \operatorname{cis} \frac{3\pi}{5} \\ z_2 &= \operatorname{cis} \frac{5\pi}{5} = -1 \\ z_3 &= \operatorname{cis} \frac{7\pi}{5} \\ z_4 &= \operatorname{cis} \frac{9\pi}{5} \end{aligned}$$

References